The Weil Conjecture. II

Pierre Deligne
English translation

(1.6) Around Jacobson–Morosov

Proposition (1.6.1). Let N be a nilpotent endomorphism of an object V of an abelian category. There exists a unique finite increasing filtration M of V such that

$$NM_i \subset M_{i-2} \quad \text{and} \quad N^k : \mathrm{Gr}^M_k V \xrightarrow{\sim} \mathrm{Gr}^M_{-k} V$$

for every $k \geq 0$.

We prove existence; uniqueness is proved in the same way. Induct on an integer d such that $N^{d+1} = 0$. For $d = 0$, take the trivial filtration, with $\mathrm{Gr}^M_i V = 0$ for $i \neq 0$. For the induction step, put

$$M_d = V, \qquad M_{d-1} = \mathrm{Ker}\, N^d, \qquad M_{-d} = \mathrm{Im}\, N^d,$$

and choose M_{-d-1} so that $\mathrm{Gr}^M_d V$ and $\mathrm{Gr}^M_{-d} V$ are respectively the coimage and the image of N^d. On $\mathrm{Ker}\, N^d / \mathrm{Im}\, N^d$, the $(d-1)$-st power of N is zero. By induction this quotient has its filtration; for $d-1 > i > -d$, define M_i in V as the inverse image in $\mathrm{Ker}\, N^d$ of the corresponding subobject of $\mathrm{Ker}\, N^d / \mathrm{Im}\, N^d$.

Remark (1.6.2). The characterization (1.6.1) shows compatibility of M with passage to the dual category.

Definition (1.6.3). The primitive part $P_i(V)$, or simply P_i, of $\mathrm{Gr}^M_i V$ is the kernel of the map induced by N:

$$N : \mathrm{Gr}^M_i V \longrightarrow \mathrm{Gr}^M_{i-2} V.$$

(1.6.4). If $i > 0$, the map $N : \mathrm{Gr}^M_i V \to \mathrm{Gr}^M_{i-2} V$ is injective, since $N^{i-1}N$ is an isomorphism; hence $P_i = 0$. For $i \geq 0$, writing that

$$N \circ N^{i+1} : \mathrm{Gr}^M_{i+2} V \longrightarrow \mathrm{Gr}^M_{-i} V \longrightarrow \mathrm{Gr}^M_{-i-2} V$$

is an isomorphism shows that $\mathrm{Gr}^M_{-i} V$ is the direct sum of P_{-i} and of the image of $\mathrm{Gr}^M_{i+2} V$ by N^{i+1}; the morphism N induces an isomorphism from this image onto $\mathrm{Gr}^M_{-i-2} V$. Repeating this construction gives, by decreasing induction on $i \geq 0$, an isomorphism of $\mathrm{Gr}^M_{-i} V$ with the sum of the P_{-j} for $j \geq i$ and $j \equiv i \pmod 2$. Using also the isomorphisms $N^i : \mathrm{Gr}^M_i V \to \mathrm{Gr}^M_{-i} V$, one obtains

$$\mathrm{Gr}^M_i V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i\ (2)}} P_{-j}. \tag{1.6.4.1}$$

Via these isomorphisms, the map $N : \mathrm{Gr}^M_i V \to \mathrm{Gr}^M_{i-2} V$ is the evident inclusion for $i > 1$ and the evident projection for $i - 2 < -1$.

Lemma (1.6.5). The morphism

$$N : (V, M) \longrightarrow (V, M \text{ shifted by } 2)$$

is strictly compatible with the filtrations.

It is enough to prove that $NM_{i+2} \supset \mathrm{Im}(N) \cap M_i$. If $i < 0$, the morphism $N : M_{i+2} \to M_i$ has surjective associated graded, hence is surjective, and $NM_{i+2} = M_i$. If $i + 2 > 0$, the morphism $N : V/M_{i+2} \to V/M_i$ has injective associated graded, hence is injective, and $N^{-1}M_i \subset M_{i+2}$.

Corollary (1.6.6). The inclusion $\mathrm{Ker}\, N \subset V$ induces isomorphisms

$$\mathrm{Gr}_i^M(\mathrm{Ker}\, N) \xrightarrow{\sim} P_i.$$

(1.6.7). Suppose now that V is a finite-dimensional vector space over a field k. First consider the case where, in a basis adapted to N, the matrix of N is one Jordan block

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}. \tag{1.6.7.1}$$

Put $\dim V = d + 1$ and index the basis vectors by the integers, differing by 2, from d down to $-d$, so that $Ne_i = e_{i-2}$ for $i \neq -d$ and $Ne_{-d} = 0$. Then M_i is spanned by the e_j with $j \leq i$.

In general, V is a sum of subspaces V_a stable under N, with N of the preceding type on V_a, and the filtration M of V is the sum of the filtrations just described on the V_a.

(1.6.8). If k has characteristic zero, the filtration M may be interpreted through the Jacobson–Morosov theorem. If

$$u : SL_2 \longrightarrow GL(V)$$

is chosen so that its lowering operator is N, and if V_j is the subspace on which $\begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}$ acts by multiplication by x^j, then M_i is the sum of the V_j for $j \leq i$. It is enough to check this for an irreducible representation of SL_2, i.e. for a symmetric power of the standard representation, where one is in the situation of (1.6.7).

This interpretation gives the behavior of M under tensor products and under passage to duals; for duals one may also invoke (1.6.2).

Proposition (1.6.9). Define the tensor product (V, N) of (V', N') and (V'', N'') by

$$V = V' \otimes V'', \qquad N = N' \otimes 1 + 1 \otimes N'',$$

and the dual of (V', N') as $((V')^\vee, -{}^tN')$.

(i) If k has characteristic zero, the filtration M of a tensor product is the tensor product of the filtrations M of the factors:

$$M_i(V' \otimes V'') = \sum_{i'+i''=i} M_{i'}(V') \otimes M_{i''}(V'').$$

(ii) The filtration M of a dual is the dual filtration:

$$M_i((V')^\vee) = M_{-i-1}(V')^\perp.$$

(iii) Under these hypotheses, formation of Gr^M is therefore compatible with tensor products and with passage to duals.

(1.6.10). Assume from now on that V is finite-dimensional over a field of characteristic zero. On $\mathrm{Gr}^M(V)$ there exists a unique representation v of SL_2 such that the differential of the standard nilpotent element is the map

$$N : \mathrm{Gr}_i^M V \longrightarrow \mathrm{Gr}_{i-2}^M V,$$

and such that $\begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}$ acts by multiplication by x^j on $\mathrm{Gr}_j^M(V)$. Existence is obtained by choosing u as in (1.6.8), identifying $\mathrm{Gr}^M(V)$ with V through the splitting $\bigoplus V_j$, and transporting u to the associated graded. Uniqueness follows from the usual $\mathfrak{sl}_2$ lemma. Formation of v is compatible with tensor products and with duals as in (1.6.9).

(1.6.11). Choose once and for all, for each integer $d \geq 0$, an irreducible representation S^d of SL_2 of dimension $d+1$, and a lowest-weight vector $e_{-d} \in S^d$. Put

$$P(d, d') = \{j \mid |d - d'| \leq j \leq d + d', \quad j - d + d' \equiv 0 \pmod 2\},$$

and choose isomorphisms of representations

$$S^d \otimes S^{d'} \simeq \bigoplus_{j \in P(d,d')} S^j, \tag{1.6.11.2}$$

$$(S^d)^\vee \simeq S^d. \tag{1.6.11.3}$$

For V, N, M and P as above, there is a unique isomorphism of SL_2-representations

$$\bigoplus_j S^j \otimes P_{-j} \xrightarrow{\sim} \mathrm{Gr}^M(V), \tag{1.6.11.4}$$

where SL_2 acts trivially on P_{-j} at the source, and by (1.6.10) at the target, such that the composite

$$P_{-j} \longrightarrow S^j \otimes P_{-j} \longrightarrow \mathrm{Gr}_{-j}^M(V)$$

using the vector e_{-j} is the identity inclusion of P_{-j}. This gives isomorphisms

$$P_{-j} \simeq \mathrm{Hom}_{SL_2}(S^j, \mathrm{Gr}^M(V)). \tag{1.6.11.5}$$

(1.6.12). Let now V' and V'' be as in (1.6.10), let $V = V' \otimes V''$, and write P_j, P'_j, P''_j for the primitive parts of V, V', V''. Combining (1.6.11.4) for V' and V'' with (1.6.11.2) gives isomorphisms, depending only on the choices (1.6.11.2),

$$P_{-j} \simeq \bigoplus_{j \in P(j', j'')} P'_{-j'} \otimes P''_{-j''}. \tag{1.6.12.1}$$

For duals one similarly obtains isomorphisms, depending only on (1.6.11.3),

$$P_{-j}((V')^\vee) \simeq P_{-j}(V')^\vee. \tag{1.6.12.2}$$

Proposition (1.6.13). Let V be an object of an abelian category, endowed with a finite increasing filtration W, and let N be a nilpotent endomorphism of V preserving W. There is at most one finite increasing filtration M of V such that

$$NM_i \subset M_{i-2} \quad \text{and} \quad N^k : \mathrm{Gr}_{i+k}^M \mathrm{Gr}_i^W V \xrightarrow{\sim} \mathrm{Gr}_{i-k}^M \mathrm{Gr}_i^W V$$

for all i, k.

The proof is by induction on the length of W. We may suppose that $W_0 = V$ and that the assertion is known for W_{-1} with the induced filtration and induced endomorphism. A simultaneous shift of W and M preserves the condition. If M satisfies the condition, its restriction to W_{-1} is determined, and the quotient filtration is the filtration (1.6.1) of $\mathrm{Gr}_0^W V$. Let c be such that $\mathrm{Gr}_i^M \mathrm{Gr}_0^W V$ is nonzero only for $-c \leq i \leq c$. For $i \geq 0$ one has

1) if $i > c$, then $M_{-i} = M_{-i}(W_{-1})$;

2) $M_{-i} = M_{-i}(W_{-1}) + N^i M_i$;

3) $M_i = \mathrm{Ker}(N^{i+1} : V \to V/M_{-i-2})$.

These identities determine M uniquely by descending induction on $|i|$.

(1.6.14). In applications the abelian category will be vector spaces over a field k of characteristic zero. We shall not have merely one nilpotent endomorphism N of V, but a nilpotent action of a one-dimensional Lie algebra $\mathfrak{n}$. The preceding theory applies as soon as one chooses $N \neq 0$ in $\mathfrak{n}$. In (1.6.13), the existence of the filtration M, and the filtration itself if it exists, are independent of the choice of N. The isomorphism N^k from $\mathrm{Gr}^M_{i+k} \mathrm{Gr}^W_i V$ to $\mathrm{Gr}^M_{i-k} \mathrm{Gr}^W_i V$ has as invariant form the canonical isomorphism

$$\mathrm{Gr}^M_{i+k} \mathrm{Gr}^W_i V(k) \xrightarrow{\sim} \mathrm{Gr}^M_{i-k} \mathrm{Gr}^W_i V, \tag{1.6.14.1}$$

where $Y(k)$ denotes $Y \otimes \mathfrak{n}^{\otimes k}$.

In the situation of (1.6.1)–(1.6.12), the filtration M and the P_j are independent of the choice of N; the isomorphisms (1.6.4.1) and (1.6.12.1–2) become

$$\mathrm{Gr}^M_i(\mathrm{Ker}\, N) \simeq P_i, \tag{1.6.14.2}$$

$$\mathrm{Gr}^M_i V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i (2)}} P_{-j}\Big(\frac{i+j}{2}\Big), \tag{1.6.14.3}$$

$$P_{-j} \simeq \bigoplus_{j \in P(j',j'')} P'_{-j'} \otimes P''_{-j''}\Big(\frac{j-j'-j''}{2}\Big), \tag{1.6.14.4}$$

$$P_{-j}((V')^\vee) \simeq P_{-j}(V')^\vee(j). \tag{1.6.14.5}$$

To remember these formulas: if Gr^M_i and P_j are regarded as of weight i, and $\mathfrak{n}$ as of weight -2, then the two sides are isobaric of the same weight. We shall also use a sheaf-theoretic version of (1.6.14).

(1.7) Local monodromy

(1.7.1). Let R be a henselian discrete valuation ring, K its field of fractions, $k = R/\mathfrak{m}$ its residue field, $\overline{K}$ an algebraic closure of K, and $\overline{k}$ the residue field of the normalization of R in $\overline{K}$. Then $\overline{k}$ is an algebraic closure of k. It is often convenient first to choose $\overline{k}$, and then $\overline{K}$, by taking an algebraic closure $\overline{R}$ of R and then an algebraic closure of the field of fractions of the strict henselization.

The inertia group is the kernel

$$0 \longrightarrow I \longrightarrow \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{Gal}(\overline{k}/k) \longrightarrow 0. \tag{1.7.1.1}$$

It, or its image in a representation, is also called the local monodromy group. If ℓ is different from the residue characteristic, there is an exact sequence

$$0 \longrightarrow P \longrightarrow I \xrightarrow{t_\ell} \widehat{\mathbb{Z}}^{(\ell')}(1) \longrightarrow 0,$$

where P is a pro-p-group and $\widehat{\mathbb{Z}}^{(\ell')}(1) = \prod_{r \neq p} \mathbb{Z}_r(1)$ is the tame inertia group. Taking the ℓ-component gives

$$t_\ell : I \longrightarrow \mathbb{Z}_\ell(1),$$

whose kernel is a profinite group of order prime to ℓ.

(1.7.2). Let $\rho : I \to GL(V)$ be an ℓ-adic representation of I. Grothendieck proved that the following hypothesis is often satisfied in practice:

> $(*)$ the restriction of ρ to a subgroup I_1 of finite index in I is unipotent.

If this holds, there is a unique nilpotent representation $\overline{\rho}$ of $\overline{\mathbb{Q}}_\ell(1)$, viewed as a one-dimensional Lie algebra, in V, such that

$$\rho(\sigma) = \exp(\overline{\rho}(t_\ell(\sigma))) \qquad (\sigma \in I_1). \tag{1.7.2.1}$$

We call $\overline{\rho}$ the logarithm of the unipotent part of the local monodromy. Equivalently it is a nilpotent morphism

$$N : V(1) \longrightarrow V,$$

and (1.7.2.1) is written

$$\rho(\sigma) = \exp(N\, t_\ell(\sigma)) \qquad (\sigma \in I_1 \subset I). \tag{1.7.2.2}$$

The theory (1.6.14–15) applies. The twist (k) is simply a Tate twist. The filtrations (1.6.1) and (1.6.13) of V will be called respectively the local monodromy filtration of V and the local monodromy filtration of V relative to W.

(1.7.3). Suppose the residue field has q elements. Define the Weil group $W(\overline{K}/K)$ as the inverse image in $\mathrm{Gal}(\overline{K}/K)$ of $W(\overline{k}/k)$. If V is an ℓ-adic representation of $W(\overline{K}/K)$, Grothendieck's theorem makes $(*)$ automatic. Hence

$$N : V(1) \longrightarrow V$$

is defined and commutes with the action of $W(\overline{K}/K)$.

Lemma (1.7.4). If F' and F'' are two liftings in $W(\overline{K}/K)$ of $F \in W(\overline{k}/k)$, then the eigenvalues of F' and F'' coincide up to multiplication by roots of unity.

It suffices to replace V by its semisimplification. Then the action of I factors through a finite quotient. There is an integer $n > 0$ such that $(F')^n$ and $(F'')^n$ have the same image in $GL(V)$, and the assertion follows. Thus the ι-weights of the eigenvalues of a lifting are well defined; one may speak of a pure, mixed, or ι-pure representation.

Proposition–Definition (1.7.5). Suppose the ι-weights of V are integral. There is then a unique finite increasing filtration M of V, stable under $W(\overline{K}/K)$, such that $\mathrm{Gr}_i^M(V)$ is ι-pure of weight i. This is the filtration by weights. Moreover $NM_i(1) \subset M_{i-2}$.

Choose a lifting F' of Frobenius in $W(\overline{K}/K)$. Let V_i' be the sum of the generalized eigenspaces of F' corresponding to eigenvalues α with $w_q(\alpha) = i$, and put $M_i' = \prod_{j \le i} V_j'$. We must show that M' is independent of F'. If M'' is obtained from another lifting F'', choose I_1 as in (1.7.2.1). Since I/I_1 is finite, there are $n > 0$ and λ such that $(F')^n \equiv (F'')^n \bmod I_1$, and in $GL(V)$

$$(F')^n = \exp(\lambda N)(F'')^n, \qquad (F')^n = \exp(\mu N)(F'')^n \exp(\mu N)^{-1}, \quad \mu = \lambda/(1-q^n).$$

Since V_i' is still the sum of the generalized eigenspaces of $(F')^n$ of the corresponding weights, the automorphism $\exp(\mu N)$ carries M' onto M''. Since N preserves M', they are equal.

Corollary (1.7.6). If V is ι-pure, then the action of I factors through a finite quotient.

By twisting one reduces to integral weight. The weight filtration has only one step. Since $NM_i \subset M_{i-2}$, one has $N = 0$, and the assertion follows.

Remark (1.7.7). The proof of (1.7.5) shows that every representation V has a unique decomposition

$$V = \sum_{\alpha} V^{\alpha}$$

where α runs through $\overline{\mathbb{Q}}_{\ell}^{\times}$ modulo roots of unity, and such that the eigenvalues of Frobenius liftings on V^{α} lie in the class α. Each V^{α} is obtained from a mixed representation by twisting.

Variant (1.7.8). If D is a smooth divisor in a regular scheme X, the preceding constructions have analogues for sheaves on $X - D$ which are tamely ramified along D. More generally, let D be a divisor with normal crossings on X, the transverse union of smooth divisors D_i $(i \in I)$. Locally choose equations $t_i = 0$. Let E be their intersection. For n invertible on X, set

$$X_n = X[t_i^{1/n}]_{i \in I}.$$

The cover $X_n \to X$ is etale over $X - D$ and totally ramified over E; above E it has the section $t_i^{1/n} = 0$. Restriction to E of a sheaf on X_n means inverse image by that section. One obtains an action of $(\mu_n)^I$ on X_n by letting $(\zeta_i)_{i \in I}$ act on $t_i^{1/n}$ by multiplication by ζ_i.

Let L be a set of primes invertible on X, and let $\mathcal{F}$ be a locally constant sheaf of sets on $X - D$, tamely ramified along D and even L-ramified along D. Abhyankar's lemma gives an integer n, with prime factors in L, such that the inverse image of $\mathcal{F}$ on $\pi^{-1}(X - D) \subset X_n$ is the restriction of a locally constant sheaf $\overline{\mathcal{F}}$ on X_n. We write $\mathcal{F}[E]$ for its restriction to E. It is equipped with an action of

$$\mathbb{Z}_L(1)^I = \prod_{\ell \in L} \mathbb{Z}_{\ell}(1)^I.$$

This construction is functorial and passes to constructible $\overline{\mathbb{Q}}_{\ell}$-sheaves, and to Weil sheaves, by passage to the limit. If X is of finite type over $\mathbb{Z}$, the action is quasi-unipotent; there are commuting nilpotent endomorphisms

$$N_i \in \mathrm{End}(\mathcal{F}[E])(-1)$$

such that an element σ in a sufficiently small open subgroup acts by $\exp(\sum_i N_i \sigma_i)$.

(1.7.9). Near an intersection of p of the D_i, D is the union of p smooth divisors, and the constructions can be repeated. If $J \subset I$, let D_J be the intersection of the D_j $(j \in J)$, let X_J be the complement of the union of the D_i for $i \notin J$, and put $D^i_J mes = D_J \cap X_J$. The divisors D_i with $i \notin J$ cut out on D_J a normal-crossing divisor with complement $D^i_J mes$, while the D_j with $j \in J$ cut out on X_J a normal-crossing divisor with complement $X - D$.

Applying (1.7.8) gives sheaves $\mathcal{F}[D_J]$ and, for $J \subset K \subset I$, canonical identifications

$$\mathcal{F}[D_J][D^i mes_{J \cup K}] \simeq \mathcal{F}[D_{J \cup K}], \tag{1.7.9.1}$$

compatible with the actions of the corresponding inertia groups. The transitivity square is

$$\begin{tikzcd}
\mathcal{F}[D_J][D_K][D_L] \arrow[r, equal] \arrow[d, equal] & \mathcal{F}[D_K][D_L] \arrow[d, equal] \\
\mathcal{F}[D_J][D_L] \arrow[r, equal] & \mathcal{F}[D_L].
\end{tikzcd}$$

(1.7.10). The constructions (1.7.8) depend on the choice of the coordinates t_i. To make them canonical one introduces the normal bundle T of E in X and the subbundles T_i tangent to D_i. The sheaf $\mathcal{F}[E]$ is replaced by a sheaf $\mathcal{F}[E]'$ on $T - \bigcup T_i$; a system of equations t_i gives a section of $T - \bigcup T_i$, and the former $\mathcal{F}[E]$ is the inverse image of $\mathcal{F}[E]'$ by that section. The construction is compatible with smooth base change. The N_i and the inertia actions act on $\mathcal{F}[E]'$, which plays the role of a restriction of $\mathcal{F}$ to a tubular neighborhood of E in X.

(1.7.11). Suppose X henselian local, $X = \mathrm{Spec}(R)$, and take for L the set of all primes invertible on X. The preceding construction has a Galois interpretation. With notation as in the source, a locally constant sheaf on $X - D$, tamely ramified along D, becomes trivial after adjoining the n-th roots of the local equations. This gives an extension

$$0 \longrightarrow \mathbb{Z}_L(1)^I \longrightarrow \pi_1^{\mathrm{mod}}(X - D, \mathrm{Spec}\,\overline{K}) \longrightarrow \mathrm{Gal}(\overline{k}/k) \longrightarrow 0,$$

and the choice of the t_i splits it. The construction of $\mathcal{F}[E]$ is precisely the corresponding sheaf on $\mathrm{Spec}(k)$ with its action of $\mathbb{Z}_L(1)^I$.

(1.7.12). If k is finite, define $W^{\mathrm{mod}}(X - D, \mathrm{Spec}\,\overline{K})$ as the inverse image of $W(\overline{k}/k)$ in the preceding tame fundamental group. The analogues of (1.7.3)–(1.7.7) hold:

(1.7.12.1) The restriction to $\mathbb{Z}_L(1)^I$ of an ℓ-adic representation of $W^{\mathrm{mod}}(X - D, \mathrm{Spec}\,\overline{K})$ is quasi-unipotent.

(1.7.12.2) If F' and F'' are two liftings of $F \in W(\overline{k}/k)$, then their eigenvalues coincide up to roots of unity; this defines the ι-weights.

(1.7.12.3) If the ι-weights of V are integral, there is a unique weight filtration M on V, as in (1.7.5), and the analogues of (1.7.6) and (1.7.7) hold.

(1.8) Local monodromy of pure sheaves

The conventions of (0.7) are in force. Let X_0 be a smooth absolutely irreducible curve over $\mathbb{F}_q$ and let $j : U_0 \hookrightarrow X_0$ be an open subset with finite complement S_0.

Lemma (1.8.1). If $\mathcal{F}_0$ is a lisse sheaf on U_0, pointwise ι-pure of weight β, then, for every $x \in |S_0|$ and every eigenvalue α of F_x on $j_*\mathcal{F}_0$, one has

$$w_{N(x)}(\alpha) \le \beta.$$

After removing from X_0 a point of U_0, one reduces to the affine case. Then

$$\prod_{x\in|U_0|} \det(1-F_x t^{\deg x}, \mathcal{F}_0)^{-1} \prod_{x\in|S_0|} \det(1-F_x t^{\deg x}, j_*\mathcal{F}_0)^{-1} = \frac{\det(1-Ft, H^1_c(X, j_*\mathcal{F}))}{\det(1-Ft, H^2_c(X, j_*\mathcal{F}))}. \quad (1.8.1.1)$$

In this formula: the first factor on the left converges for $|t| < q^{-(\beta+2)/2}$ and has neither zeros nor poles there; by (1.4.3), the right hand side has no pole in the same region. Comparing, the remaining factors have no pole in that region. Hence, if α is an eigenvalue of F_x on $j_*\mathcal{F}_0$,

$$w_{N(x)}(\alpha) \le \beta + 2. \quad (1.8.1.2)$$

Applying this to tensor powers $\mathcal{F}_0^{\otimes k}$, and using that $j_*(\mathcal{F}_0^{\otimes k})$ contains $(j_*\mathcal{F}_0)^{\otimes k}$ as a subsheaf, gives $k w_{N(x)}(\alpha) \le k\beta + 2$, and letting $k \to \infty$ proves the lemma.

Remark (1.8.2). Comparing zeros in (1.8.1.1), the numerator cannot vanish in the same region. Thus, for an eigenvalue α of F on $H^1_c(X, j_*\mathcal{F})$, one obtains $w_q(\alpha) \le \beta + 2$.

(1.8.3). Let $\overline{s}$ be a geometric point over $s \in S_0$, let η be the generic point of the henselian trait with finite residue field $X_{0,(s)}$, and let $\overline{\eta}$ be a geometric generic point of its strict henselization. The fibre $\mathcal{F}_{0,\overline{\eta}}$ is a representation of the Weil group $W(\overline{\eta}/\eta)$. Let M be its local monodromy filtration.

Theorem (1.8.4). With the notation of (1.8.1) and (1.8.3), if $\mathcal{F}_0$ is pointwise ι-pure of weight β, then

$$\mathrm{Gr}_i^M(\mathcal{F}_{0,\overline{\eta}})$$

is ι-pure of weight $\beta + i$ as a representation of $W(\overline{\eta}/\eta)$.

By twisting (1.2.7), reduce to $\beta = 0$. After replacing X_0 by a finite cover, and replacing U_0, $\mathcal{F}_0$, and s by their inverse images, assume that the local monodromy is unipotent on all of inertia:

$$\rho(\sigma) = \exp(t_\ell(\sigma)N) \qquad (\sigma \in I).$$

Then $\operatorname{Ker} N = (j_*\mathcal{F}_0)_{\overline{s}}$. By (1.8.1), eigenvalues of Frobenius on $\operatorname{Ker} N$ have absolute value at most 1. If α is an eigenvalue of Frobenius on $P_{-j}(\mathcal{F}_{0,\overline{\eta}})$, then (1.6.14.2) gives $|\iota(\alpha)| \leq 1$. Applying this to $\mathcal{F}_0 \otimes \mathcal{F}_0$, and using (1.6.14.4), shows that $\alpha^2 q^j$ occurs on $P_0(\mathcal{F}_{0,\overline{\eta}} \otimes \mathcal{F}_{0,\overline{\eta}})$; hence $|\iota(\alpha^2 q^j)| \leq 1$ and $|\iota(\alpha)| \leq q^{-j/2}$. Applying the same argument to the dual sheaf, and using (1.6.14.5), gives the opposite inequality. Thus the eigenvalues on P_{-j} have absolute value $q^{-j/2}$; (1.6.14.3) gives the theorem.

Corollary (1.8.5). Suppose that $\mathcal{F}_0$ is a lisse sheaf on U_0 and admits a finite increasing filtration W by lisse subsheaves such that each $\mathrm{Gr}_i^W(\mathcal{F}_0)$ is pointwise ι-pure of weight i. Let W also denote the filtration induced on $\mathcal{F}_{0,\overline{\eta}}$. Then the local monodromy filtration of $V = \mathcal{F}_{0,\overline{\eta}}$ relative to W exists, and it agrees with the weight filtration of $\mathcal{F}_{0,\overline{\eta}}$ of (1.7.5).

Indeed, by (1.7.5) and (1.8.4) applied to the $\mathrm{Gr}_i^W(\mathcal{F}_0)$, the characteristic conditions (1.6.13) are satisfied by the weight filtration.

(1.8.6). Let D_0 be a smooth divisor in a smooth scheme X_0 over $\mathbb{F}_q$, let $U_0 = X_0 - D_0$, and let $\mathcal{F}_0$ be a lisse sheaf on U_0, tamely ramified along D_0. Suppose that $\mathcal{F}_0$ has a finite increasing filtration W by lisse subsheaves such that $\mathrm{Gr}_i^W \mathcal{F}_0$ is pointwise ι-pure of weight i. If D_0 is defined by an equation t, let $\mathcal{F}_0[D_0]$ be the sheaf on D_0 with its action of $\mathbb{Z}_\ell(1)$ obtained by (1.7.8). Restricting to curves transverse to D_0, one obtains from (1.8.5):

Corollary (1.8.7). With the notation of (1.8.6), the local monodromy filtration around D of $\mathcal{F}_0[D_0]$, relative to W, exists. For this filtration M, the object $\mathrm{Gr}_i^M \mathcal{F}_0[D_0]$ is pointwise ι-pure of weight i.

Remark (1.8.8). With the notation of (1.8.1), if $\mathcal{F}_0$ is lisse on U_0, then $j_*\mathcal{F}_0$ is the invariant part of the local monodromy. Under the hypotheses of (1.8.4), $\operatorname{Ker} N \subset M_0$, so the filtration induced on $(j_*\mathcal{F}_0)_s$ has Gr_i^M pure of weight $\beta + i$ and is zero for $i > 0$. The analogous statement for a smooth divisor is obtained from (1.8.7). Under the hypotheses of (1.8.6), W and monodromy define a filtration M on $j_*\mathcal{F}_0|_{D_0}$ for which $\mathrm{Gr}_i^M(j_*\mathcal{F}_0)|_{D_0}$ is lisse and ι-pure of weight i.

Corollary (1.8.9). Let X_0 be of finite type over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be an open subset, and let $\mathcal{F}_0$ be an ι-mixed lisse sheaf of pointwise weights $\leq \beta$ on U_0. Then $j_*\mathcal{F}_0$ is again ι-mixed of pointwise weights $\leq \beta$. If the pointwise weights of $\mathcal{F}_0$ are integral, those of $j_*\mathcal{F}_0$ are integral as well.

The proof uses the standard devissages: devissage in $\mathcal{F}_0$, suppression of a locally closed subscheme, normalization after finite surjective maps, making infinity a divisor, and localization at the generic point of infinity. These reductions reduce the assertion to (1.8.8).

Corollary (1.8.10). Let $\mathcal{F}_0$ be a lisse sheaf on X_0 of finite type over $\mathbb{F}_q$, and let $j : U_0 \hookrightarrow X_0$ be a dense open subset. If the restriction of $\mathcal{F}_0$ to U_0 is pointwise ι-pure, then $\mathcal{F}_0$ itself is pointwise ι-pure.

Indeed, if β is the weight on U_0, then $\mathcal{F}_0 \to j_*j^*\mathcal{F}_0$ and the same map for the dual show, using (1.8.9), the two inequalities on the pointwise weights.

Corollary (1.8.11). Every lisse ι-mixed sheaf on a normal scheme X_0 of finite type over $\mathbb{F}_q$ is an iterated extension of pointwise ι-pure lisse sheaves.

One reduces to the irreducible case. Since X_0 is normal, the restriction to a suitable nonempty open is still irreducible, hence pointwise pure after shrinking; one applies (1.8.10). Later (3.4.1) will remove the normality hypothesis.

Corollary (1.8.12). Let $\mathcal{F}_0$ be a lisse ι-mixed sheaf on a connected scheme X_0 of finite type over $\mathbb{F}_q$. If $\mathcal{F}_0$ is ι-pure at one point of X_0, then it is pointwise ι-pure.

For normal X_0 this follows from (1.8.11), since weights are read on every fibre. In general, apply the result to the inverse image on the normalization: the locus where the sheaf is pure of a fixed weight is open and closed.

Variant (1.8.13). In this paragraph one may replace ι-pure and ι-mixed by pure and mixed, and also by ι-pure of integral weights and ι-mixed of integral weights.

(1.8.14). In the heuristic dictionary of [2], I, Theorem (1.8.4) corresponds to Schmid's theory of the asymptotic behavior of polarizable variations of Hodge structure on the punctured disk D^*. Corollary (1.8.5) suggests the following problem.

Call a "variation of mixed Hodge structures" on a smooth analytic space S the data of a local system $V_{\mathbb{Z}}$ of free finitely generated $\mathbb{Z}$-modules, a finite increasing filtration W of $V_{\mathbb{Q}} = V_{\mathbb{Z}} \otimes \mathbb{Q}$ by local subsystems, and a finite decreasing filtration F of $V = V_{\mathbb{Z}} \otimes \mathcal{O}$ by vector subbundles, varying holomorphically, satisfying transversality $\nabla F^p \subset \Omega^1 \otimes F^{p-1}$ and such that W and F define a mixed Hodge structure at every point of S.

Variations arising in algebraic geometry have additional properties; for example, the variations of Hodge structures $\mathrm{Gr}_i^W(V)$ are polarizable. Let (V, W, F) be a variation of mixed Hodge structures on D^*, let $t \in D^*$, let V denote the fibre at t, and let $T \in \mathrm{End}(V)$ be monodromy. Assume for simplicity that T is unipotent and put $N = \log T$.

Problem (1.8.15). Find a class of "good" variations of mixed Hodge structures on D^* such that, for (V, W, F) good and T unipotent, there exists on V a finite increasing filtration M satisfying $NM_i \subset M_{i-2}$ and such that N^k induces isomorphisms

$$N^k : \mathrm{Gr}^M_{a+k} \mathrm{Gr}^W_a V \xrightarrow{\sim} \mathrm{Gr}^M_{a-k} \mathrm{Gr}^W_a V.$$

One would also like: (a) V should be asymptotic, in a suitable sense, to a nilpotent orbit; a good variation for which F is replaced by the transport to $\exp(t)$ of $\exp(-tN)F$ should be the filtration deduced from F by horizontal transport from t to $\exp(t)$; and (b) for a nilpotent orbit, (V, M, F) should still be a variation of mixed Hodge structures.